TYPE OF DATA → CLASSIFY + SYMBOLS

TRIGGER Classify data or identify symbols. **BME423 · V8.** Population = entire group; sample = measured subset. Parameters are fixed; sample statistics vary across samples. Keep full calculator precision.

x_i (observation i); n (sample size); $\bar{x}$ (sample mean); μ (population mean); s (sample SD); σ (population SD); s^2, σ^2 (variances). $\sum$ means add the specified terms.

WRITE Interval–continuous: values within a range (BP, height, membrane potential). Interval–discrete: counts/rounded values (joints, texts, age in years). Categorical–nominal: unordered labels (country, ZIP). Categorical–ordered: ranked levels (cancer stage, severity). **TRAP** Numeric labels such as ZIP are not interval measurements.

ν (degrees of freedom); α (test significance level); f (mass/density); F (CDF).

RAW LIST → MEAN, VARIANCE, SD, SEM

TRIGGER A raw list; asks for center, spread or error in its estimated mean. **WORKED** Heights (inches): 65, 69, 59, 64, 62, 67, 68, 62, 64, 65. Find mean, variance, SD and SEM.

1. Add the observations; divide by their count

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$n = 10, \quad \sum_{}^{n} x_i = 645$$

$$\bar{x} = \frac{645}{10} = 64.5$$

2. Subtract the mean, then square each difference

$$d_i = x_i - \bar{x}, \qquad d_i^2 = (x_i - \bar{x})^2$$

$$d_1 = 65 - 64.5 = .5, \quad d_1^2 = .5^2 = .25$$
$$d_2 = 69 - 64.5 = 4.5, \quad d_2^2 = 4.5^2 = 20.25$$
$$d_3 = 59 - 64.5 = -5.5, \quad d_3^2 = (-5.5)^2 = 30.25$$
$$d_4 = 64 - 64.5 = -.5, \quad d_4^2 = (-.5)^2 = .25$$
$$d_5 = 62 - 64.5 = -2.5, \quad d_5^2 = (-2.5)^2 = 6.25$$
$$d_6 = 67 - 64.5 = 2.5, \quad d_6^2 = 2.5^2 = 6.25$$
$$d_7 = 68 - 64.5 = 3.5, \quad d_7^2 = 3.5^2 = 12.25$$
$$d_8 = 62 - 64.5 = -2.5, \quad d_8^2 = (-2.5)^2 = 6.25$$
$$d_9 = 64 - 64.5 = -.5, \quad d_9^2 = (-.5)^2 = .25$$
$$d_{10} = 65 - 64.5 = .5, \quad d_{10}^2 = .5^2 = .25$$

$$\sum d_i^2 = 82.5$$

3. Divide by one less than the sample size

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{82.5}{10 - 1} = 9.16667$$

4. Square-root variance; scale SD for a mean

$$s = \sqrt{s^2}, \qquad SEM = \frac{s}{\sqrt{n}}$$

$$s = \sqrt{9.16667} = 3.02765$$

$$SEM = \frac{3.02765}{\sqrt{10}} = .957427$$

WRITE Estimated mean 64.5 inches; individual SD 3.02765 inches; mean-estimation error .957427 inch. **TRAP** SD describes individuals; SEM describes sample means. Variance has squared units.

TRAP A complete population uses N and its population mean; a sample variance divides by $n - 1$.

$$\mu = \frac{\sum_i x_i}{N}, \qquad \sigma^2 = \frac{\sum_i (x_i - \mu)^2}{N}$$

Squared-sum check for the same variance numerator

$$\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - \frac{(\sum_i x_i)^2}{n}$$

$$\sum_i x_i^2 = 41685, \qquad (\sum_i x_i)^2 = 645^2 = 416025$$

$$41685 - \frac{416025}{10} = 41685 - 41602.5 = 82.5$$

The same 82.5 is divided by 9 for the sample variance.

Estimators: $\bar{x}$ estimates μ; s estimates σ. Unbiased = centered correctly across repeated samples. For Normal populations, the sample mean has the minimum-variance property taught in class.

$$\widehat{SE}(s^2) = s^2\sqrt{\frac{2}{n-1}} = 9.16667\sqrt{\frac{2}{9}} = 4.32121$$

This estimates uncertainty in the sample variance for Normal observations; it is different from SEM.

SORTED DATA → PERCENTILES, MEDIAN, IQR

TRIGGER A data list; asks for a percentile, median, IQR or range. **WORKED** Sorted heights: 59, 62, 62, 64, 64, 65, 65, 67, 68, 69. Find the quartiles and spread.

1. Locate a percentile between sorted ranks

$$R = 1 + \frac{p(n-1)}{100} = I + d$$

$$R_{25} = 1 + \frac{25(10-1)}{100} = 3.25$$

$$R_{50} = 1 + \frac{50(10-1)}{100} = 5.5$$

$$R_{75} = 1 + \frac{75(10-1)}{100} = 7.75$$

p (percentile number); I (integer rank); d (fraction of the next gap). **TRAP** Rank locates a value; interpolate the measurements, not their ranks.

2. Interpolate the measured values, not the ranks

$$x_p = x_{(I)} + d[x_{(I+1)} - x_{(I)}]$$

$$Q_1 = 62 + .25(64 - 62) = 62.5$$

$$Q_2 = 64 + .5(65 - 64) = 64.5$$

$$Q_3 = 65 + .75(67 - 65) = 66.5$$

3. IQR and observed range (59–69)

$$IQR = Q_3 - Q_1, \quad \text{range} = x_{\max} - x_{\min}$$

$$IQR = 66.5 - 62.5 = 4$$

$$\text{range} = 69 - 59 = 10$$

WRITE Median 64.5; Q1 62.5; Q3 66.5; IQR 4; range 59–69. A straight Normal Q–Q plot supports Normal shape. Box = Q1–Q3 with median; usual whisker cutoff = 1.5×IQR; histogram = bin counts.

VALUES + PROBABILITIES → MEAN + VARIANCE

TRIGGER Possible values and their probabilities. **WORKED** Check validity; find mean, variance, CDF and probability from 2 inclusive to 7 exclusive.

x	-1	2	3	7
$p(x)$	.55	.15	.10	.20

1. Check both probability requirements

$$0 \le p(x) \le 1, \qquad \sum_x p(x) = 1$$

$$.55 + .15 + .10 + .20 = 1$$

2. Weight each value by its probability

$$\mu = E[X] = \sum_x x\,p(x)$$

$$\mu = (-1)(.55) + 2(.15) + 3(.10) + 7(.20) = -.55 + .30 + .30 + 1.40 = 1.45$$

X (random variable); x (value); $p(x)$ (its probability). **TRAP** Weight values by probability; average them equally only for equal probabilities.

3. Square the values before weighting

$$E[X^2] = \sum_x x^2 p(x)$$

$$E[X^2] = 1(.55) + 4(.15) + 9(.10) + 49(.20) = .55 + .60 + .90 + 9.80 = 11.85$$

4. Direct variance: weight EVERY squared deviation

$$v_x = (x - \mu)^2 p(x), \quad \text{Var}(X) = \sum_x v_x$$

$$v_{-1} = (-1 - 1.45)^2(.55) = 3.30138$$
$$v_2 = (2 - 1.45)^2(.15) = .045375$$
$$v_3 = (3 - 1.45)^2(.10) = .24025$$
$$v_7 = (7 - 1.45)^2(.20) = 6.1605$$
$$\text{Var}(X) = 3.30138 + .045375 + .24025 + 6.1605 = 9.7475$$

5. Check with the second-moment shortcut

$$\text{Var}(X) = E[X^2] - (E[X])^2$$
$$\text{Var}(X) = 11.85 - 1.45^2 = 9.7475$$
$$SD(X) = \sqrt{9.7475} = 3.1221$$

6. Add only the included masses

$$P(X \in A) = \sum_{x \in A} p(x)$$

$$P(2 \le X < 7) = p(2) + p(3) = .15 + .10 = .25$$

WRITE Mean 1.45; variance 9.7475; requested probability .25.

AT MOST / AT LEAST → DISCRETE CDF

TRIGGER "At most," "at least," or a CDF graph.

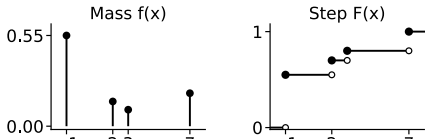
1. Accumulate probabilities through the endpoint

$$F(x) = P(X \le x) = \sum_{t \le x} p(t)$$

$$F(x) = \begin{cases} 0 & x < -1 \\ .55 & -1 \le x < 2 \\ .70 & 2 \le x < 3 \\ .80 & 3 \le x < 7 \\ 1 & x \ge 7 \end{cases}$$

2. Include or exclude the endpoint mass

$$P(X > a) = 1 - F(a)$$
$$P(X > 2) = 1 - .70 = .30$$
$$P(X \ge 2) = 1 - P(X < 2) = 1 - .55 = .45$$
$$P(-1 < X < 3) = p(2) = .15$$



WRITE / SKETCH Mass: separate spikes. CDF: jump by each mass; flat between values. **TRAP** Fill each jump's upper point; open lower point. A discrete point can have positive probability.

LOADED DIE → USE THE STATED MASSES

TRIGGER A die has unequal stated probabilities. **WORKED** $p(1) = .20$; each of 2–6 has .16. Find $P(X \le 3)$, mean and variance.

Weight using the stated masses

$$P(X \le a) = \sum_{x \le a} p(x), \quad E[X] = \sum xp(x)$$

$$P(X \le 3) = .20 + .16 + .16 = .52$$
$$E[X] = 1(.20) + (2 + 3 + 4 + 5 + 6)(.16) = 3.4$$
$$E[X^2] = 1(.20) + (4 + 9 + 16 + 25 + 36)(.16) = 14.6$$
$$\text{Var}(X) = E[X^2] - \mu^2 = 14.6 - 3.4^2 = 3.04$$

WRITE Probability .52; mean 3.4; variance 3.04. **TRAP** A loaded die does not give every face probability 1/6.

SUCCESSIVE EVENTS → UPDATE WHAT REMAINS

TRIGGER Successive draws; check whether outcomes are replaced. **WORKED** Draw two cards from 52 without replacement. Find the chance both are kings.

Without replacement: update the second draw

$$P(A \cap B) = P(A)P(B \mid A)$$

$$P(\text{two kings}) = \frac{4}{52}\frac{3}{51} = .00452489$$

TRAP Without replacement, update numerator and denominator; independent probabilities stay unchanged. **WRITE** Two kings: .00452489. Three fair flips: exactly two heads = HHT/HTH/THH, 3 of 8 equally likely sequences.

CONTINUOUS DENSITY → INTEGRATE THE AREA

TRIGGER A density formula over an interval; asks for areas or moments. **WORKED** $f(x) = x/2$ on (0,2), zero elsewhere. Find area, CDF, mean, variance, tail and median.

Integration rule: dx names the integration variable. Raise its power by one, divide by that new power; evaluate upper bound minus lower bound.

$$\int_a^b kx^r\,dx = \left[\frac{kx^{r+1}}{r+1}\right]_a^b \quad (r \ne -1)$$

1. Check total area and nonnegative height

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^2 \frac{x}{2}\,dx = \frac{1}{2}\left[\frac{x^{1+1}}{1+1}\right]_0^2 = \left[\frac{x^2}{4}\right]_0^2 = \frac{2^2}{4} - \frac{0^2}{4} = 1$$

TRAP Density height is not probability; area is probability. The density must be nonnegative. $\int_a^b g(x)\,dx = G(b) - G(a)$ means upper-bound value minus lower-bound value.

2. Accumulate area only up to the requested value

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$F(x) = \int_0^x \frac{t}{2}\,dt = \left[\frac{t^2}{4}\right]_0^x = \frac{x^2}{4}$$

$$F(x) = \begin{cases} 0 & x \le 0 \\ x^2/4 & 0 < x < 2 \\ 1 & x \ge 2 \end{cases}$$

3. Multiply by the outcome before integrating

$$\mu = E[X] = \int x f(x)\,dx$$

$$E[X] = \int_0^2 x\frac{x}{2}\,dx = \int_0^2 \frac{x^2}{2}\,dx = \frac{1}{2}\left[\frac{x^{2+1}}{2+1}\right]_0^2 = \left[\frac{x^3}{6}\right]_0^2 = \frac{2^3}{6} - \frac{0^3}{6} = \frac{8}{6} = \frac{4}{3}$$

4. Use a second moment for variance

$$E[X^2] = \int x^2 f(x)\,dx, \quad \sigma^2 = E[X^2] - \mu^2$$

$$E[X^2] = \frac{1}{2}\int_0^2 x^3\,dx = \frac{1}{2}\left[\frac{x^{3+1}}{3+1}\right]_0^2 = \left[\frac{x^4}{8}\right]_0^2 = \frac{2^4}{8} - \frac{0^4}{8} = 2$$

$$\sigma^2 = 2 - \left(\frac{4}{3}\right)^2 = \frac{18 - 16}{9} = \frac{2}{9}$$

5. A right tail is the unaccumulated area

$$P(X > a) = 1 - F(a)$$

$$F(1.5) = \frac{1.5^2}{4} = .5625$$

$$P(X > 1.5) = 1 - .5625 = .4375$$

$$= \int_{1.5}^2 \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_{1.5}^2 = \frac{4}{4} - \frac{2.25}{4}$$

6. A median splits the total area in half

$$F(m) = .5$$

$$\frac{m^2}{4} = .5 \Rightarrow m^2 = 2 \Rightarrow m = \sqrt{2}$$

WRITE Mean 4/3, variance 2/9, median $\sqrt{2}$. **TRAP** Keep the root inside the support. Continuous points have probability zero; inclusive endpoints do not change an area.

DENSITY CHANGES FORMULA → PIECEWISE CDF

TRIGGER The density uses different formulas in different intervals. **WORKED** $f(x) = x$ on (0,1), $2 - x$ on [1,2). Find its CDF and $P(.5 < X < 1.5)$.

Carry the completed first area into the second piece

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$0 < x < 1: \quad F(x) = \int_0^x t\,dt = \frac{x^2}{2}$$

$$1 \le x < 2: \quad F(x) = \frac{1}{2} + \int_1^x (2 - t)\,dt$$

$$= \frac{1}{2} + \left[2t - \frac{t^2}{2}\right]_1^x = \frac{1}{2} + \left(2x - \frac{x^2}{2}\right) - \left(2 - \frac{1}{2}\right)$$

$$= 2x - \frac{x^2}{2} - 1$$

$$F(x) = 0\ (x \le 0), \qquad F(x) = 1\ (x \ge 2)$$

Subtract accumulated areas

$$P(a < X < b) = F(b) - F(a)$$

$$F(.5) = \frac{.5^2}{2} = .125$$

$$F(1.5) = 2(1.5) - \frac{1.5^2}{2} - 1 = .875$$

$$P(.5 < X < 1.5) = .875 - .125 = .75$$

WRITE The requested interval probability is .75. **TRAP** Carry earlier area into the next piece; do not restart the CDF at zero. The CDF stays continuous at 1.

UNKNOWN ENDPOINT → TOTAL AREA ONE

TRIGGER Unknown support endpoint; the total density area must be one. **WORKED** $f(x) = .1 + x$ on $0 < x \le c$, zero elsewhere. Find c, CDF, moments and probabilities.

1. Force the full support area to equal one

$$\int_0^c f(x)\,dx = 1$$

$$\int_0^c (.1 + x)\,dx = \left[.1x + \frac{x^2}{2}\right]_0^c = 1$$

$$.1c + \frac{c^2}{2} = 1$$

$$2(.1c) + 2\left(\frac{c^2}{2}\right) = 2(1)$$

$$.2c + c^2 = 2 \Rightarrow c^2 + .2c - 2 = 0$$

2. Solve the quadratic; keep the valid endpoint

$$Ac^2 + Bc + D = 0: \quad c = \frac{-B \pm \sqrt{B^2 - 4AD}}{2A}$$

$$c = \frac{-.2 \pm \sqrt{.2^2 - 4(1)(-2)}}{2}$$

$$= \frac{-.2 \pm \sqrt{8.04}}{2}$$

$$\sqrt{8.04} = 2.83549, \quad c = \frac{-.2 + 2.83549}{2} = 1.31774$$

$A = 1, B = .2, D = -2$. **TRAP** Use the root inside the support; the negative root $c_- = -1.51774$ is invalid here.

WRITE Upper endpoint $c = 1.31774$.

3. Integrate only to the requested value

$$F(x) = \int_0^x f(t)\,dt$$

$$F(x) = \left[.1t + \frac{t^2}{2}\right]_0^x = .1x + \frac{x^2}{2}$$

$$F(x) = 0\ (x \le 0), \qquad F(x) = 1\ (x \ge c)$$

4. Multiply by x and expand before integrating

$$E[X] = \int_0^c x f(x)\,dx$$

$$x(.1 + x) = .1x + x^2$$

$$E[X] = \int_0^c (.1x + x^2)\,dx$$

$$= \left[.05x^2 + \frac{x^3}{3}\right]_0^c$$

$$= .05(1.31774)^2 + \frac{(1.31774)^3}{3}$$

$$= .849556$$

5. Square x, not the entire density

$$E[X^2] = \int_0^c x^2 f(x)\,dx$$

$$x^2(.1 + x) = .1x^2 + x^3$$

$$E[X^2] = \left[\frac{x^3}{30} + \frac{x^4}{4}\right]_0^c$$

$$= \frac{(1.31774)^3}{30} + \frac{(1.31774)^4}{4}$$

$$= .830089$$

6. Subtract squared mean from second moment

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$\mathrm{Var}(X) = .830089 - (.849556)^2$$

$$= .108344$$

7. Use the CDF and complement

$$P(X > a) = 1 - F(a), \quad P(X < b) = F(b)$$

$$F(.5) = .1(.5) + \frac{.5^2}{2} = .05 + .125 = .175$$

$$P(X > .5) = 1 - F(.5) = 1 - .175 = .825$$

$$P(X < .6) = F(.6) = .1(.6) + \frac{.6^2}{2} = .06 + .18 = .24$$

$$P(X = .5) = 0$$

8. Transform the mean without a new density

$$E[aX + b] = aE[X] + b$$

$$E[2X + 1] = 2(.849556) + 1 = 2.69911$$

CONSTANT DENSITY → UNIFORM

TRIGGER Constant density across a fixed range. **WORKED** For Uniform(0,1), find height, probability below .5, mean and variance.

Spread total area one across the interval

$$f(x) = \frac{1}{b-a}, \quad P(u < X < v) = \frac{v-u}{b-a}$$

$$f(x) = \frac{1}{1-0} = 1, \qquad P(0 < X < .5) = \frac{.5-0}{1-0} = .5$$

$$E[X] = \int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

$$E[X] = \frac{a+b}{2} = \frac{0+1}{2} = .5$$

$$E[X^2] = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

$$\mathrm{Var}(X) = E[X^2] - \mu^2 = \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12}$$

$$= \frac{1}{12} = \frac{(b-a)^2}{12}$$

Unknown coefficient: factor it out of the area

$$f(x) = kx\ (0 < x < 2), \quad k\int_0^2 x\,dx = 1$$

$$k\left[\frac{x^2}{2}\right]_0^2 = 1 \Rightarrow 2k = 1 \Rightarrow k = \frac{1}{2}$$

WRITE Mean .5; variance 1/12. **TRAP** Normalize over the entire support. A CDF is zero before its support and one after it.

NORMAL CUTOFFS → LEFT / RIGHT / BETWEEN

TRIGGER Normal mean and SD plus measurement cutoffs; asks for probability. **WORKED** Glucose $X \sim N(95, 8.7^2)$. Find the areas below 85, between 85 and 115, and above 115.

Write the Normal density and accumulated area

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$f_X(x) = \frac{1}{8.7\sqrt{2\pi}} \exp\left[-\frac{(x-95)^2}{2(8.7)^2}\right]$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

TRAP The second Normal parameter is variance; 8.7 is SD here. Standardize and use the table; the Normal CDF integral need not be solved by hand.

1. Convert measurement cutoffs to SD units

$$z = \frac{x - \mu}{\sigma}$$

$$z_{85} = \frac{85 - 95}{8.7} = -1.14943 \approx -1.15$$

$$z_{115} = \frac{115 - 95}{8.7} = 2.29885 \approx 2.30$$

2. Read accumulated LEFT area in the Normal table

$$\Phi(z) = P(Z \le z), \quad \Phi(-z) = 1 - \Phi(z)$$

$$\Phi(1.15) = .8749 \Rightarrow \Phi(-1.15) = .1251$$

$$\Phi(2.30) = .9893$$

TABLE The supplied table repeats complete z / LEFT-area pairs. Find 1.15 → .8749; 2.30 → .9893. For negative z, subtract the positive-z area from one.

3. Use the CDF for each requested region

$$P(X < a) = \Phi(z_a)$$

$$P(X < 85) = \Phi(-1.15) = .1251$$

$$P(a < X < b) = \Phi(z_b) - \Phi(z_a)$$

$$P(85 < X < 115) = .9893 - .1251 = .8642$$

$$P(X > b) = 1 - \Phi(z_b)$$

$$P(X > 115) = 1 - .9893 = .0107$$

TI-84: DISTR → normalcdf(lower,upper,95,8.7). Use −1E99/1E99 for infinite bounds. Calculator values are .12519, .864053 and .0107567; small table-rounding differences are expected.

$$z_0 = \frac{0 - 95}{8.7} = -10.9195$$

$$P(X < 0) = \Phi(-10.9195) \approx 0, \quad P(X = 95) = 0$$

WRITE Table probabilities: .1251, .8642, .0107; they sum to one. An exact continuous point has zero area.

GIVEN NORMAL AREA → CUTOFF / PERCENTILE

TRIGGER Given a Normal percentile, top-tail fraction or middle range. **WORKED** Find glucose Q1, Q3 and the central 95% interval.

1. Match cumulative area to a standard cutoff

$$\Phi(z_p) = p, \qquad x_p = \mu + z_p\sigma$$

$$z_{.25} = -.67449, \quad z_{.75} = +.67449$$

$$x_{25} = 95 + (-.67449)(8.7) = 89.1319$$

$$x_{75} = 95 + (.67449)(8.7) = 100.868$$

2. Put half the omitted area in each tail

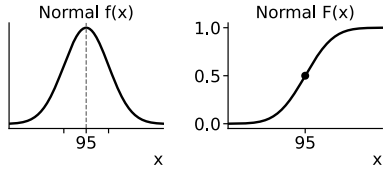
$$P(-z_* < Z < z_*) = .95 \Rightarrow \Phi(z_*) = .975$$

$$z_* = 1.96$$

$$x_L = \mu - z_*\sigma = 95 - 1.96(8.7) = 77.948$$

$$x_U = x_{97.5} = \mu + z_*\sigma = 95 + 1.96(8.7) = 112.052$$

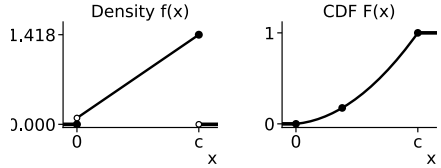
TABLE / TI-84 invNorm(left area,95,8.7). Cumulative .975 gives z = 1.96. **TRAP** Right .025 means left .975; enter .975, not 97.5.



WRITE Q1 89.1319; Q3 100.868; central 95% 77.948–112.052 mg/dL. **SKETCH** Bell centered at μ; shade the requested area. Normal CDF rises smoothly 0→1, crosses .5 at μ, and has no jumps. Larger SD widens the bell.

ASKED TO DRAW → CONTINUOUS PDF + CDF

TRIGGER Asked to draw a continuous density and its accumulated probability.



WRITE / PDF $f(x) = .1 + x$: zero outside (0,c]; rising line inside, from .1 to .1+c. Shade intervals underneath. **TRAP** $f(.5) = .6$ is a height, not probability.

WRITE / CDF Zero before 0; $F(x) = .1x + x^2/2$ inside; flat at 1 after c. It is continuous and increasing; its slope is density height. Formula changes can create corners, not point-mass jumps.

AVERAGE OF n → SAMPLING DISTRIBUTION / SEM

TRIGGER Probability concerns an average of n observations. **WORKED** Individuals have mean 4 and SD 1.5; independent n=50. Find the chance the sample mean is 3.5–3.8.

1. Keep the center; reduce variance for averaging

$$E[\bar{X}] = \mu, \quad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$E[\bar{X}] = 4, \quad \mathrm{Var}(\bar{X}) = \frac{1.5^2}{50} = .045$$

$$SD(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{50}} = .212132$$

$\bar{X}$ (random sample mean); $\bar{x}$ (one observed mean). If individuals are Normal, means are Normal at any sample size. Otherwise the CLT gives an approximately Normal mean for sufficiently large independent samples (course rule: $n \ge 30$).

2. Divide each mean cutoff by the MEAN spread

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$z_L = \frac{3.5 - 4}{.212132} = -2.35702$$

$$z_U = \frac{3.8 - 4}{.212132} = -.942809$$

3. Subtract the accumulated areas

$$P(a < \bar{X} < b) = \Phi(z_b) - \Phi(z_a)$$

$$\Phi(-.942809) = .172889$$

$$\Phi(-2.35702) = .00921106$$

$$P(3.5 < \bar{X} < 3.8) = .172889 - .00921106$$

$$= .163678$$

TABLE Round z to −.94 and −2.36; .1736−.0091=.1645. **WRITE** About 16.4% of sample averages fall in the interval. **TRAP** Use SEM for means; use σ for an individual, without dividing by $\sqrt{n}$.

SHIFT / SCALE / SUM → MEAN + VARIANCE

TRIGGER A constant is added, a variable is scaled, or independent variables are combined. **WORKED** For the glucose variable, find $E[X - 26]$ and its variance.

1. Apply coefficients directly to the mean

$$E[aX + b] = a\mu + b, \quad \mathrm{Var}(aX + b) = a^2\sigma^2$$

$$E[X - 26] = 95 - 26 = 69$$

$$\mathrm{Var}(X - 26) = 1^2(8.7^2) = 75.69$$

2. For independent variables, ADD variances

$$E[aX + bY] = a\mu_X + b\mu_Y$$

$$\mathrm{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2$$

For two independent measurements with the glucose distribution:

$$E[X + Y] = 95 + 95 = 190$$

$$\mathrm{Var}(X + Y) = 8.7^2 + 8.7^2 = 151.38$$

$$\mathrm{Var}(X - Y) = 8.7^2 + (-1)^2(8.7^2) = 151.38$$

WRITE A shift changes the mean, not spread. **TRAP** Square coefficients for variance; even subtraction adds independent variances. $X + X = 2X$ is not an independent sum; its variance is $4\sigma^2$.

$$E[h(X)] = \sum_x h(x)p(x) \quad \text{(discrete)}$$

$$E[h(X)] = \int h(x)f(x)\,dx \quad \text{(continuous)}$$

LARGER SAMPLE → SMALLER SEM

TRIGGER Asked how changing sample size changes error in the estimated mean. **WORKED** Height SD 3.02765, SEM .957427, n=10. Find SEM for four times as many observations.

Compare sample sizes with individual spread held fixed

$$SEM_{\text{new}} = SEM_{\text{old}}\sqrt{\frac{n_{\text{old}}}{n_{\text{new}}}}$$

$$n_{\text{new}} = 4(10) = 40$$

$$SEM_{\text{new}} = .957427\sqrt{\frac{10}{40}} = .478714$$

WRITE Four times n halves SEM to .478714. For target SEM, $n = (s/SEM_{\text{target}})^2$; round upward. **TRAP** A more precise mean does not require smaller individual variation.

ONE EVENT COUNT → PROPORTION + SE

TRIGGER One group; asks for the event proportion and its error. **WORKED** 8 events among 61 patients. Estimate the proportion and SE.

Use event count divided by all patients

$$\hat{p} = \frac{x}{n}, \quad s_{\text{binary}} = \sqrt{\hat{p}(1-\hat{p})}$$

$$\hat{p} = \frac{8}{61} = .131148$$

$$s_{\text{binary}} = \sqrt{.131148(.868852)} = .337562$$

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$SE(\hat{p}) = \sqrt{\frac{.131148(.868852)}{61}} = .0432203$$

WRITE Estimated event proportion .131148; SE .0432203. **TRAP** SE describes uncertainty in the estimated proportion, not variation among individual zero/one outcomes.

SUPPLIED NORMAL OUTPUT → AREA OR CUTOFF?

TRIGGER Supplied Normal command/output; decide whether input is a cutoff or an area.

WRITE pnorm: cutoff → area; qnorm: area → cutoff. Default LEFT; lower.tail=FALSE means RIGHT.

```
pnorm(85,95,8.7)
[1] 0.12519
pnorm(115,95,8.7,lower.tail=FALSE)
[1] 0.0107567
qnorm(.975,95,8.7)
[1] 112.052
```

TRAP Right .025 equals left .975. Third argument is SD; use SEM for a sample average.

ONE SAMPLE vs FIXED MEAN → t TEST

TRIGGER One numerical sample versus a fixed reference mean. **WORKED** USC students: n=16, mean LDL 103, SD 16 mg/dL; reference 115. Independent Normal responses: is the student population mean different at .05?

1. State the population hypotheses

$$H_0 : \mu = \mu_0, \qquad H_A : \mu \neq \mu_0$$
$$H_0 : \mu = 115, \qquad H_A : \mu \neq 115$$

In words: the undergraduate population mean equals 115 under the null; it differs from 115 under the alternative. μ_0 is the fixed reference, not another sample mean.

2. Estimate the mean's standard error

$$SE = \frac{s}{\sqrt{n}}$$
$$SE = \frac{16}{\sqrt{16}} = \frac{16}{4} = 4$$

3. Standardize the observed departure

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$t = \frac{103 - 115}{4} = \frac{-12}{4} = -3$$

4. Use the one-sample degrees of freedom

$$\nu = n - 1, \qquad |t| > t_{\text{crit}} \Rightarrow \text{reject } H_0$$
$$\nu = 16 - 1 = 15$$
$$t_{\text{crit}} = t_{.025,15} = 2.13145$$
$$|t| = 3 > 2.13145$$

TABLE t row 15, total two-tail .05 → 2.131. **WRITE** At .05, student population mean LDL differs from 115 mg/dL; the observed mean is lower.

p-value $p = 2P(T_{15} \geq 3) = .00897274$. **TRAP** The reference is fixed, not another sample. Compare p with .05; p is not the probability the null is true.

TWO NUMERICAL GROUPS → POOLED t

TRIGGER Two independent numerical groups; common variance, possibly unequal n. **WORKED** Control n=22, mean 49, SD 8.4; intervention n=27, mean 57, SD 9.5. Normal populations: do their means differ at .05?

1. State the hypotheses for the two populations

$$H_0 : \mu_1 = \mu_2, \qquad H_A : \mu_1 \neq \mu_2$$

In words: control and intervention have equal population means under the null; their population means differ under the alternative.

2. Pool VARIANCES, weighted by their df

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$s_1^2 = 8.4^2 = 70.56, \qquad s_2^2 = 9.5^2 = 90.25$$
$$s_p^2 = \frac{21(70.56) + 26(90.25)}{22 + 27 - 2}$$
$$= \frac{1481.76 + 2346.50}{47} = 81.4523$$

3. Compute the standard error of the difference

$$SE_{\text{diff}} = \sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$
$$SE_{\text{diff}} = \sqrt{81.4523\left(\frac{1}{22} + \frac{1}{27}\right)}$$
$$= 2.59213$$

4. Divide the observed mean gap by its SE

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE_{\text{diff}}}$$
$$t = \frac{49 - 57}{2.59213} = -3.08627$$

5. Compare with a two-sided cutoff

$$\nu = n_1 + n_2 - 2, \quad |t| > t_{\text{crit}}$$
$$\nu = 22 + 27 - 2 = 47$$
$$|t| = 3.08627 > t_{.025,47} = 2.01174$$

TABLE df 47 lies between rows 46/48; total two-tail .05 cutoffs 2.013/2.011. The statistic exceeds both. **WRITE** Population means differ at .05; intervention's observed mean is higher (p=.00339345). **TRAP** Weight variances by df when n differs; do not just average them.

SUPPLIED t OUTPUT → READ df + p

TRIGGER A printed t-test result; asks for the test, df, p or conclusion.

```
Welch Two Sample t-test
t = -4.5298, df = 15.611
p-value = 0.0003625
mean of x   mean of y
64.5        69.7
```

WRITE At .05, population mean heights differ; the first observed mean is lower. Read t (statistic), df and p. **TRAP** Welch can have fractional df: use the printed df, not pooled df.

```
Two Sample t-test
t = -4.5298, df = 18
p-value = 0.0002594
```

TABLE / METHOD "Two Sample t-test" uses pooled $\nu = n_1 + n_2 - 2$. The hand method assumes common variance; follow the supplied test name.

EQUAL-SIZE NUMERICAL GROUPS → ANOVA

TRIGGER Several numerical groups of equal size; asks whether any population mean differs. **WORKED** Filgrastim: n=15 per dose; neutrophils in 10^9 cells/L. Test all-equal means at .05, then compare pairs.

Dose	Mean	SD
Low	3.7	1.1
Medium	4.6	1.4
High	5.2	2.1

1. State the overall population hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3$$
$$H_A : \text{not all population means are equal}$$

In words: the three dose population means are equal under the null; at least one differs under the alternative. Assume independent Normal responses and equal population variances.

2. Average the group variances for within spread

$$s_{\text{wit}}^2 = \frac{\sum_{j=1}^{m} s_j^2}{m}$$
$$s_{\text{wit}}^2 = \frac{1.1^2 + 1.4^2 + 2.1^2}{3}$$
$$= \frac{1.21 + 1.96 + 4.41}{3} = 2.52667$$

$m = 3$ (number of groups); $n = 15$ (people per group); s_j (SD in group j).

3. Average the means; square their deviations

$$\bar{\bar{x}} = \frac{\sum_j \bar{x}_j}{m}$$
$$\bar{\bar{x}} = \frac{3.7 + 4.6 + 5.2}{3} = 4.5$$
$$\sum_j (\bar{x}_j - \bar{\bar{x}})^2 = (3.7 - 4.5)^2$$
$$+ (4.6 - 4.5)^2 + (5.2 - 4.5)^2$$
$$= .64 + .01 + .49 = 1.14$$

4. Divide by m−1, then multiply by group size

$$s_{\bar{x}}^2 = \frac{\sum_j (\bar{x}_j - \bar{\bar{x}})^2}{m - 1}, \quad s_{\text{bet}}^2 = n s_{\bar{x}}^2$$
$$s_{\bar{x}}^2 = \frac{1.14}{3 - 1} = .57$$
$$s_{\text{bet}}^2 = 15(.57) = 8.55$$

5. Divide between by within

$$F = \frac{s_{\text{bet}}^2}{s_{\text{wit}}^2}$$
$$F = \frac{8.55}{2.52667} = 3.38391$$

6. Use numerator and denominator df in that order

$$\nu_1 = m - 1, \quad \nu_2 = m(n - 1)$$
$$\nu_1 = 3 - 1 = 2, \quad \nu_2 = 3(15 - 1) = 42$$
$$F = 3.38391 > F_{.05;2,42} = 3.21994$$

TABLE F: numerator df 2 across top, denominator df 42 down side; upper .05 lightface → 3.22. **WRITE** At .05, at least one dose population mean differs (p=.0433947). **TRAP** Use variances, not SDs; the overall test does not identify the pair.

WHICH MEAN PAIRS? → BONFERRONI

TRIGGER Named mean pairs; asks for comparisons at one family significance level. **WORKED** Compare the filgrastim dose pairs at family $\alpha = .05$.

1. State the hypothesis for EACH dose pair

$$H_{0,ij} : \mu_i = \mu_j, \quad H_{A,ij} : \mu_i \neq \mu_j$$

In words: each null says the two named dose populations have equal means; each alternative says those population means differ.

2. Divide family alpha by the requested pair count

$$k = \begin{cases} m(m-1)/2 & \text{all pairs} \\ m - 1 & \text{vs control} \end{cases}$$
$$\alpha_{\text{pair}} = \frac{\alpha_{\text{family}}}{k}$$
$$k = \frac{3(3-1)}{2} = 3, \quad \alpha_{\text{pair}} = \frac{.05}{3} = .0166667$$
$$\nu = m(n - 1) = 3(14) = 42$$
$$t_{\text{crit}} = t_{\alpha_{\text{pair}}/2,42} = 2.49367$$

TRAP Count the requested family, not every possible pair. Retain ANOVA within variance and residual df. Total two-tail .0166667 equals one-tail .00833333.

TABLE / TI-84 t row 42, total two-tail .0166667. Missing column: DISTR → invT; enter LEFT area and original df:

$$t_{\text{crit}} = \text{invT}(1 - \alpha_{\text{pair}}/2, \nu)$$
$$= \text{invT}(1 - (.05/3)/2, 42) = 2.49367$$

3. Compute the shared SE using the original within variance

$$SE_{ij} = \sqrt{s_{\text{wit}}^2\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$
$$SE_{ij} = \sqrt{2.52667\left(\frac{1}{15} + \frac{1}{15}\right)}$$
$$= \sqrt{\frac{2(2.52667)}{15}} = .580421$$

4. Low versus Medium

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{SE_{ij}}$$
$$|t_{LM}| = \frac{|3.7 - 4.6|}{.580421} = 1.5506$$
$$1.5506 < 2.49367 \Rightarrow \text{do not reject } H_{0,LM}$$

Remaining dose pairs continue at the top of the next column.

BONFERRONI → REMAINING DOSE PAIRS

TRIGGER Continue the same three-pair family with the original SE and cutoff.

$$SE_{ij} = .580421, \quad t_{\text{crit}} = 2.49367$$

5. Low versus High

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{SE_{ij}}$$
$$|t_{LH}| = \frac{|3.7 - 5.2|}{.580421} = 2.58433$$
$$2.58433 > 2.49367 \Rightarrow \text{reject } H_{0,LH}$$

6. Medium versus High

$$|t_{ij}| = \frac{|\bar{x}_i - \bar{x}_j|}{SE_{ij}}$$
$$|t_{MH}| = \frac{|4.6 - 5.2|}{.580421} = 1.03373$$
$$1.03373 < 2.49367 \Rightarrow \text{do not reject } H_{0,MH}$$

WRITE At family .05, only low versus high population means differ. **TRAP** Nonsignificant pairs do not establish identical means.

ORDERED t PAIRS → HOLM; STOP AT FAILURE

TRIGGER Supplied t pairs; asks which differ while controlling one family. **WORKED** Five G1 comparisons; df 46; ordered $|t|$ = 3.3, 2.9, 2.3, 2.2, 2.1. Use family .05.

$$H_{0,1j} : \mu_1 = \mu_j, \quad H_{A,1j} : \mu_1 \neq \mu_j$$

In words: each null says G1 and that named group have equal population means; its alternative says their means differ.

1. Order strongest evidence first; adjust the remaining family

$$\alpha_j = \frac{\alpha_{\text{family}}}{k - j + 1}$$
$$\alpha_1 = \frac{.05}{5} = .0100, \quad \alpha_2 = \frac{.05}{4} = .0125$$
$$\alpha_3 = \frac{.05}{3} = .0166667$$

TRAP Order p smallest first; with common df, order $|t|$ largest first. Stop at the first failure.

2. Check each ordered pair using its own two-sided cutoff

$$|t_j| > t_{\alpha_j/2,\nu} \Rightarrow \text{reject and continue}$$
$$G1 : G6 : \quad 3.3 > t_{.005,46} = 2.687 \quad \text{reject}$$
$$G1 : G3 : \quad 2.9 > t_{.00625,46} = 2.59973 \quad \text{reject}$$
$$G1 : G5 : \quad 2.3 < t_{.00833333,46} = 2.48469 \quad \text{STOP}$$

WRITE At family .05, G1 differs from G6 and G3. Retain G1:G5, G1:G4 and G1:G2 without further tests. **TABLE** t row 46, current two-tail alpha; if absent, invT(1−alpha/2,46).

PAIRS vs CONTROL → HOLM CHI-SQUARE

TRIGGER Printed chi-square pairs; asks which treatments differ from control. **WORKED** Mortality statistics: 18.2341, 9.49443, 4.68951; each df 1. Test the three control comparisons at family .05.

1. Order the evidence, then reduce the remaining family

$$\alpha_j = \frac{\alpha}{k - j + 1}$$
$$k = 3 : \quad \alpha_1 = .05/3 = .0166667$$
$$\alpha_2 = .05/2 = .025, \qquad \alpha_3 = .05/1 = .05$$

Order p smallest first; with common df, order χ^2 largest first. **In words:** each null says treatment and control have equal mortality probabilities; its alternative says they differ.

2. Compare each ordered statistic with its own cutoff

$$\chi_j^2 > \chi_{\alpha_j,1}^2 \Rightarrow \text{reject } H_{0,j}$$
$$18.2341 > 6.635 > \chi_{.0166667,1}^2 \quad \text{reject}$$

TABLE Chi-square row 1: .0166667 is absent; the stricter .01 cutoff 6.635 already proves the first rejection.

$$9.49443 > \chi_{.025,1}^2 = 5.02389$$
$$4.68951 > \chi_{.05,1}^2 = 3.84146$$

WRITE All three treatments differ from control at family .05. **TRAP** Only the three requested control pairs count; stop at the first failure and retain all remaining nulls.

CRITICAL VALUE / p RANGE → READ THE TABLE

TRIGGER Asked for a critical value, p-value range or missing table entry.

TABLE z: complete z / adjacent LEFT area; negatives use symmetry. t: df row, full TWO-TAILED alpha (.05 means .05). F: numerator df across top, denominator down side; UPPER tail. Chi-square: df row, RIGHT tail; either supplied chi-square table works.

Bound a p-value between printed table columns

$$t_{\text{two tail .10,4}} < |t| < t_{\text{two tail .05,4}}$$
$$2.132 < 2.58630 < 2.776 \Rightarrow .05 < p_{\text{two sided}} < .10$$

WRITE Here .05 < p < .10. Missing df? Read neighboring rows: above both cutoffs → reject; below both → do not reject; between them → exact cutoff or supplied p needed.

TRAP / TI-84 Positive t cutoff $\text{invT}(1 - \alpha/2, df)$; two-sided p $2\,\text{tcdf}(|t|, 1E99, df)$. Halve alpha for invT only, not again for the two-tail table.

SUPPLIED ANOVA → READ F + WITHIN VARIANCE

TRIGGER Printed ANOVA output; asks for an overall decision or follow-up pairs.

```
            Df Sum Sq Mean Sq F value Pr(>F)
Group        2  779.4   389.7   5.832 0.0106 *
Residuals   19 1269.5    66.8
```

Read: Group df 2 = F numerator df; Residuals df 19 = denominator and pairwise df. Residuals Mean Sq 66.8 = within variance; F value 5.832 = test statistic; Pr(>F) .0106 = p-value.

$$F = \frac{MS_{\text{Group}}}{MS_{\text{Residual}}} = 5.832 \quad \text{(unrounded inputs)}$$

WRITE At .05, at least one population mean differs; at .01, a difference is not established. Retain residual variance and df for requested pairs, using their actual sample sizes.

If residual STANDARD ERROR is supplied, square it

$$s_{\text{wit}}^2 = (\text{residual standard error})^2$$
$$s_{\text{wit}}^2 = (8.17407)^2 = 66.8154 \approx 66.8$$

TRAP Residual standard error is an SD: square it. Residual Mean Sq is already a variance.

TWO PROPORTIONS → YATES z

TRIGGER Two groups; each person has event/no event. **WORKED** Thrombi: 18/25 placebo versus 6/19 aspirin. Are population event probabilities different at .05?

1. State the population hypotheses and sample fractions

$$H_0 : \pi_1 = \pi_2, \quad H_A : \pi_1 \neq \pi_2$$
$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{18}{25} = .72$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{6}{19} = .315789$$

In words: placebo and aspirin have equal thrombus probabilities under the null; the probabilities differ under the alternative. x_i is the event count; n_i is that group's total.

2. Pool event counts for the null probability

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$
$$\hat{p} = \frac{18 + 6}{25 + 19} = \frac{24}{44} = .545455$$

3. Calculate the null SE and half-count adjustment

$$SE_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$
$$SE_0 = \sqrt{.545455(.454545)\left(\frac{1}{25} + \frac{1}{19}\right)}$$
$$= .151547$$
$$c_Y = \frac{1}{2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$
$$c_Y = \frac{1}{2}\left(\frac{1}{25} + \frac{1}{19}\right) = .0463158$$

4. Shrink the absolute observed gap, then standardize

$$|z_Y| = \frac{\max(0, |\hat{p}_1 - \hat{p}_2| - c_Y)}{SE_0}$$
$$D_Y = \max(0, |.72 - .315789| - .0463158)$$
$$= .357895$$
$$|z_Y| = \frac{.357895}{.151547} = 2.36161$$

5. Make the two-sided decision

$$|z_Y| > z_{\alpha/2} \Rightarrow \text{reject } H_0$$
$$2.36161 > 1.95996$$

TABLE Normal cumulative .975, or t infinite-df row, total two-tail .05 → 1.960. **WRITE** Population thrombus probabilities differ at .05; aspirin's observed rate is lower. **TRAP** Retain Yates for the course's 2×2 comparisons.

ROUNDED PROPORTIONS → POOL EXACT COUNTS

TRIGGER Rounded group proportions plus an exact combined event count. **WORKED** Female births after father-only dioxin: .565 of 186; mother-only: .455 of 220. Exactly 205/406 female overall. Compare population probabilities at .05.

$$H_0 : \pi_F = \pi_M, \quad H_A : \pi_F \neq \pi_M$$

In words: the two populations have equal female-birth probabilities under the null; they differ under the alternative.

1. Pool the exact combined count; keep the stated group fractions

$$\hat{p} = \frac{\text{total events}}{n_1 + n_2}$$
$$\hat{p} = \frac{205}{186 + 220} = \frac{205}{406} = .504926$$

TRAP Pool the exact total event count; do not reconstruct group counts from rounded proportions.

2. Find the null SE and half-count correction

$$SE_0 = \sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}$$
$$SE_0 = \sqrt{.504926(.495074)(1/186 + 1/220)}$$
$$= .0498017$$
$$c_Y = \tfrac{1}{2}(1/n_1 + 1/n_2)$$
$$c_Y = .5(1/186 + 1/220) = .0049609$$

3. Shrink the gap, divide by SE and compare

$$|z_Y| = \frac{\max(0, |\hat{p}_1 - \hat{p}_2| - c_Y)}{SE_0}$$
$$D_Y = |.565 - .455| - .0049609 = .105039$$
$$|z_Y| = \frac{.105039}{.0498017} = 2.10915 > 1.960$$

TABLE Normal cumulative .975 / t infinite-df two-tail .05 → 1.960. **WRITE** Female-birth probabilities differ at .05; father-exposed sample's fraction is higher.

ONE LIST vs TARGET % → GOODNESS OF FIT

TRIGGER One observed category list versus stated target percentages. **WORKED** Counts 255, 125, 90; target probabilities .60, .25, .15. Test fit at .05.

1. State the target distribution; find expected counts

$$H_0 : (\pi_1, \pi_2, \pi_3) = (.60, .25, .15)$$
$$H_A : \text{at least one population proportion differs}$$
$$N = 255 + 125 + 90 = 470, \quad E_i = Np_i$$
$$E_1 = 470(.60) = 282$$
$$E_2 = 470(.25) = 117.5, \quad E_3 = 470(.15) = 70.5$$

In words: the population follows the specified target proportions under the null; it does not follow that distribution under the alternative.

2. Work EVERY category, then sum

$$c_i = \frac{(O_i - E_i)^2}{E_i}, \quad \chi^2 = \sum_i c_i$$
$$c_1 = \frac{(255 - 282)^2}{282} = 2.58511$$
$$c_2 = \frac{(125 - 117.5)^2}{117.5} = .478723$$
$$c_3 = \frac{(90 - 70.5)^2}{70.5} = 5.39362$$
$$\chi^2 = 2.58511 + .478723 + 5.39362$$
$$= 8.45745$$

3. Use category count minus one

$$\nu = k - 1, \quad \chi^2 > \chi^2_{\alpha,\nu}$$
$$\nu = 3 - 1 = 2, \qquad 8.45744681 > 5.99146$$

TABLE Chi-square row 2, upper .05 → 5.991. **WRITE** Reject the target proportions at .05 (p=.01457). **TRAP** Use specified targets, not observed fractions; targets sum to one. No Yates.

2×2 COUNTS → YATES CHI-SQUARE

TRIGGER Two groups × two outcomes; observed COUNTS. **WORKED** Test whether folate use and neural tube defects are associated at .05, using the table below.

Group	Event	No event	Total
Folate	4	471	475
No folate	27	498	525
Total	31	969	1000

1. State independence in the population

$$H_0 : \text{group and outcome are independent}$$
$$H_A : \text{group and outcome are associated}$$

In words: the malformation probability is the same with and without folate under the null; it differs under the alternative.

2. Build ALL expected counts from margins

$$E_{ij} = \frac{R_i C_j}{N}$$
$$E_{11} = \frac{475(31)}{1000} = 14.725$$
$$E_{12} = \frac{475(969)}{1000} = 460.275$$
$$E_{21} = \frac{525(31)}{1000} = 16.275$$
$$E_{22} = \frac{525(969)}{1000} = 508.725$$

R_i (row total); C_j (column total); N (grand total); O_{ij} (observed count).

3. Calculate EVERY corrected cell contribution

$$c_{ij} = \frac{\max(0, |O_{ij} - E_{ij}| - .5)^2}{E_{ij}}$$
$$c_{11} = \frac{(|4 - 14.725| - .5)^2}{14.725} = 7.10021$$
$$c_{12} = \frac{(|471 - 460.275| - .5)^2}{460.275} = .227148$$
$$c_{21} = \frac{(|27 - 16.275| - .5)^2}{16.275} = 6.424$$
$$c_{22} = \frac{(|498 - 508.725| - .5)^2}{508.725} = .205515$$

4. Add contributions; compare on the right tail

$$\chi^2_Y = \sum_i \sum_j c_{ij}, \quad \nu = (r-1)(c-1)$$
$$\chi^2_Y = 7.10021 + .227148$$
$$+6.424 + .205515 = 13.9569$$
$$\nu = (2-1)(2-1) = 1$$
$$13.9569 > \chi^2_{.05,1} = 3.84146$$

TABLE Chi-square row 1, upper .05 → 3.841. **WRITE** Folate and neural tube defects are associated at .05 (p=.00018705). **TRAP** Use counts, not percentages; Yates shrinks the absolute gap before squaring.

LARGER COUNT TABLE → ORDINARY CHI-SQUARE

TRIGGER More than two rows or columns of observed counts. **WORKED** Depression event/no-event counts: (36,131), (41,132), (75,104). Compare the population probabilities at .05.

1. State the hypotheses; total rows and columns

$$H_0 : \pi_1 = \pi_2 = \pi_3, \quad H_A : \text{not all } \pi_i \text{ equal}$$
$$(R_1, R_2, R_3) = (167, 173, 179)$$
$$(C_1, C_2) = (152, 367), \qquad N = 519$$

In words: all three population event probabilities are equal under the null; at least one differs under the alternative.

2. Calculate ALL six null expected counts

$$E_{ij} = \frac{R_i C_j}{N}$$
$$E_{11} = \frac{167(152)}{519} = 48.9094$$
$$E_{12} = \frac{167(367)}{519} = 118.091$$
$$E_{21} = \frac{173(152)}{519} = 50.6667$$
$$E_{22} = \frac{173(367)}{519} = 122.333$$
$$E_{31} = \frac{179(152)}{519} = 52.4239$$
$$E_{32} = \frac{179(367)}{519} = 126.576$$

3. Calculate ALL six contributions (no Yates)

$$c_{ij} = \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$
$$c_{11} = \frac{(36 - 48.9094)^2}{48.9094} = 3.40739$$
$$c_{12} = \frac{(131 - 118.091)^2}{118.091} = 1.41124$$
$$c_{21} = \frac{(41 - 50.6667)^2}{50.6667} = 1.8443$$
$$c_{22} = \frac{(132 - 122.333)^2}{122.333} = .763851$$
$$c_{31} = \frac{(75 - 52.4239)^2}{52.4239} = 9.7223$$
$$c_{32} = \frac{(104 - 126.576)^2}{126.576} = 4.02667$$

4. Add and select the larger-table df

$$\chi^2 = \sum c_{ij}, \quad \nu = (r-1)(c-1)$$
$$\chi^2 = 3.40739 + 1.41124 + 1.8443$$
$$+.763851 + 9.7223 + 4.02667$$
$$= 21.1757$$
$$\nu = (3-1)(2-1) = 2$$
$$21.1757 > \chi^2_{.05,2} = 5.99146$$

TABLE Chi-square row 2, upper .05 → 5.991. **WRITE** Population depression probabilities differ at .05. **TRAP** No Yates for 3×2; follow-up pairs identify which groups differ.

```
X-squared = 21.176, df = 2
p-value = 2.522e-05
```

`2.522e-05` = .00002522

FOLLOWED GROUPS → RR, ARR, NNT / NNH

TRIGGER Followed groups; asks for treatment benefit or harm. **WORKED** Aspirin 6/19 events; placebo 18/25. Find RR, ARR and NNT.

1. Find BOTH event risks from group totals

$$r_g = \frac{\text{events in group } g}{\text{all people in group } g}$$
$$r_T = \frac{6}{19} = .315789$$
$$r_C = \frac{18}{25} = .72$$

T (treatment); C (control). **TRAP** Risk divides events by ALL group members; odds divide events by nonevents.

2. Divide for relative risk; subtract for reduction

$$RR = \frac{r_T}{r_C}, \qquad ARR = r_C - r_T$$
$$RR = \frac{6/19}{18/25} = .438596$$
$$ARR = \frac{18}{25} - \frac{6}{19} = .404211$$

WRITE Aspirin risk is .439 times placebo risk; absolute reduction 40.42 percentage points. **TRAP** Relative and absolute reductions differ.

3. Take the reciprocal of ABSOLUTE risk reduction

$$NNT = \frac{1}{ARR}$$
$$NNT = \frac{1}{.404211} = 2.47396 \Rightarrow 3$$

WRITE Treat 3 people per additional event prevented over the study interval. **TRAP** Round NNT upward; reciprocals use decimal risks.

4. If treatment increases harm, use NNH

$$ARI = r_T - r_C, \quad NNH = \frac{1}{ARI}$$
$$ARI = .08 - .03 = .05, \qquad NNH = \frac{1}{.05} = 20$$

Small baseline risks: relative benefit versus absolute benefit

$$RR = \frac{r_T}{r_C}, \quad RRR = 1 - RR, \quad ARR = r_C - r_T$$
$$r_C = 10/1000 = .010, \quad r_T = 8/1000 = .008$$
$$RR = \frac{.008}{.010} = .8, \quad RRR = 1 - .8 = .20$$
$$ARR = .010 - .008 = .002, \quad NNT = 1/.002 = 500$$
$$\frac{NNT}{NNH} = \frac{500}{20} = 25$$

WRITE Here 20% relative benefit is .2 percentage points absolute. With NNH 20, 25 additional harms accompany one additional prevented event.

RISK COMPARISON → NAME THE NUMERATOR

TRIGGER Risk in one group relative to another.

Compare no-folate risk with folate risk

$$r_g = \frac{x_g}{n_g}, \qquad RR_{A/B} = \frac{r_A}{r_B}$$
$$r_{\text{no folate}} = \frac{27}{525} = .0514286$$
$$r_{\text{folate}} = \frac{4}{475} = .00842105$$
$$RR_{\text{no/yes}} = \frac{27/525}{4/475} = 6.10714$$
$$RR_{\text{yes/no}} = \frac{4/475}{27/525} = .163743$$

WRITE No-folate risk is 6.11 times folate risk. **TRAP** Reversing the ratio gives its reciprocal; name its direction.

CASES CHOSEN BY DISEASE → ODDS RATIO

TRIGGER Selected by existing disease; compare earlier exposure. **WORKED** Cases: 50 exposed, 14 unexposed; controls: 43 exposed, 35 unexposed. Find the exposure OR.

	Exposed	Unexposed
Cases	50	14
Controls	43	35

Divide exposed by unexposed in EACH selected group

$$OR = \frac{a/b}{c/d} = \frac{ad}{bc}$$
$$O_{\text{case}} = \frac{50}{14} = 3.57143$$
$$O_{\text{control}} = \frac{43}{35} = 1.22857$$
$$OR = \frac{50/14}{43/35} = \frac{50(35)}{14(43)}$$
$$= \frac{1750}{602} = 2.90698$$

WRITE Cases have 2.91 times controls' exposure odds. **TRAP** Case–control sampling fixes case numbers; sample fractions cannot estimate population disease risks. Use OR.

ASSIGNED / SELECTED / FOLLOWED → DESIGN

TRIGGER Selection, treatment assignment or follow-up.

WRITE Randomized trial: assign treatment, follow outcomes. Cohort: follow exposure groups without assignment; both give risks/RR. Case–control: select by outcome, assess earlier exposure; OR. Parallel: different people; crossover: same people in different periods. Masking hides assignment from participants, clinicians/evaluators or statisticians. Sham mimics the procedure. **TRAP** Crossover needs stability/no carryover; case–control samples do not give disease risks.

SUPPLIED CHI-SQUARE → READ STATISTIC + p

TRIGGER Printed statistic, df or decision.

```
Pearson's Chi-squared test with
Yates' continuity correction
X-squared = 5.5772, df = 1
p-value = 0.0182
```

WRITE At .05, aspirin/placebo probabilities differ: p=.0182 < .05. X-squared = statistic; p-value = RIGHT-tail area. **TRAP** Yates is already included; do not correct again. Here $\chi^2_Y = z_Y^2$.